Sammy Al Hashemi

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(647)-463-2638

Education

- Bachelor of Applied Science and Engineering
  - Graduating: April 2019
- Department of Engineering Science
- Major in physics

cGPA: 3.56

Hardware Skills

- Verilog
- Electronic Circuit Analysis

Software Skills

- Python
- Javascript
- Angular
- nodeJS
- Java

Courses Taken

- Algorithms/Foundations of Computing
- Data Strcutures & Algorithms
- Machine Learning and Data Mining
- Electronic Circuits
- Digital Electronics

Website

www.sammyalhashemi.com

Work Experience

Device Characterization and Product Engineering Intern

Microsemi Corp.

July 2017 - August 2018

- Worked on development of test structures and designs using Microsemi's design software, Libero, to assess and characterize Microsemi FPGA products that went to market.
- Specifically, characterized the Mathblock component of Microsemi's G5 FPGA product as well as performance-related AC tests on other combinatorial and sequential elements found in the FPGA.
  - Example components include look-up tables, carry chains, digital I/O's, characterized mainly for max frequency.
  - Tested chains on different corners of the FPGA fabric.
  - AC characterization was done on Verigy 93k tester offsite using test vectors generated on site.
- Inside Libero, I created Verilog test structures, making use of JTAG to read input from test programs created using a company code.
- Wrote sections of the characterization report found in the official product manual used to facilate customer knowledge in our device.

Projects

Thesis

- Using IBM Q's python SDK, `qiskit`, I am programming scalable versions of Grover's algorithm for my thesis under Proffessor Glen Gulak.
- I will be presenting my results in April 2019.

Personal Website

- Single-page application written in an Angular6 frontend with Typescript
- Uses Firebase's Cloud Storage to store and asynchronously fetch larger files such as pictures.
- Animations used to liven up pages include:
- Uses complex angular routing features such as lazy loading
- RxJS usage for asynchronous data handling (migrated from RxJS5 to RxJS6)

Tennis Ranking Scraper

- Writing a webapp (still in development) that webscrapes the ranking history of tennis players Roger Federer, Rafael Nadal, Novak Djokovic, Andy Murray, and Nick Kyrgios.
- Front end is written in react.
- Backend webscraper is an `express` server that uses `cheerio.js` wrapped in a request-promise.